When Models Commit: Candidate Formation and Winner Isolation Across Model Depth

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Abstract

Language models produce one answer, but they need not arrive there in one step: an answer can become plausible before it becomes decisive. We study that gap in multiple-choice question answering by scoring exact option-text continuations layer by layer in nine instruction-tuned open-weight models from the Qwen, Gemma, and Llama families on 989 ARC-Challenge and CommonsenseQA questions. For each prompt, we measure when the eventual answer first enters a small plausible-answer set and when it later opens a stable margin over the runner-up. The gap is large and consistent. By 20% of normalized depth, the eventual answer is already plausible in 72% of evaluable cases, but it is already committed in only 4%; every evaluable model-dataset cell is positive under the preregistered primary rule. Two control readouts show the same broad pattern while changing which cases are measurable and, often, which answer wins. In the setting we study, candidate formation usually comes earlier than winner isolation.

1 Introduction

A multiple-choice answer hides more than one decision. A model can first narrow the field to a few answers that still seem possible and only later settle on one final winner. Standard evaluation only shows the last step. It tells us what answer came out, but not when that answer first became a serious option inside the network.

This paper asks a simple question. Does the eventual answer become one of the plausible answers before the model writes the margin that finally separates it from the runner-up? Figure 1 previews the answer. In the pooled primary analysis, by 20% of normalized depth the eventual answer is already in the plausible answer set for 72% of evaluable cases, but only 4% are already stably committed.

We answer that question in a deliberately direct way. We build a normal multiple-choice prompt, end it with the suffix "Final answer:", and then score the exact strings the model could write next. On the main path, that string is the full option text itself, such as "photosynthesis". So instead of training a separate probe to guess the answer from hidden states, we measure the scores of the answer strings that the model is actually being asked to produce.

Our setting is intentionally narrow and easy to audit. We evaluate nine instruction-tuned open-weight models from the Qwen 2.5 Instruct, Gemma 2 IT, and Llama Instruct families on 989 ARC-Challenge and CommonsenseQA questions. The claim of the paper is limited to that setting. Within it, the pattern is consistent: the eventual answer usually enters the plausible answer set well before it becomes stably isolated from the runner-up. All 18 evaluable model-dataset cells are positive under the preregistered primary rule (Figure 2, Table 2).

The paper makes three contributions. First, it gives a simple measurement framework for separating "still possible" from "finally chosen" by scoring the task's own answer strings rather than an external probe target. Second, it shows a large and consistent gap between those two stages in the pooled curves and in every evaluable ARC-Challenge and Common-

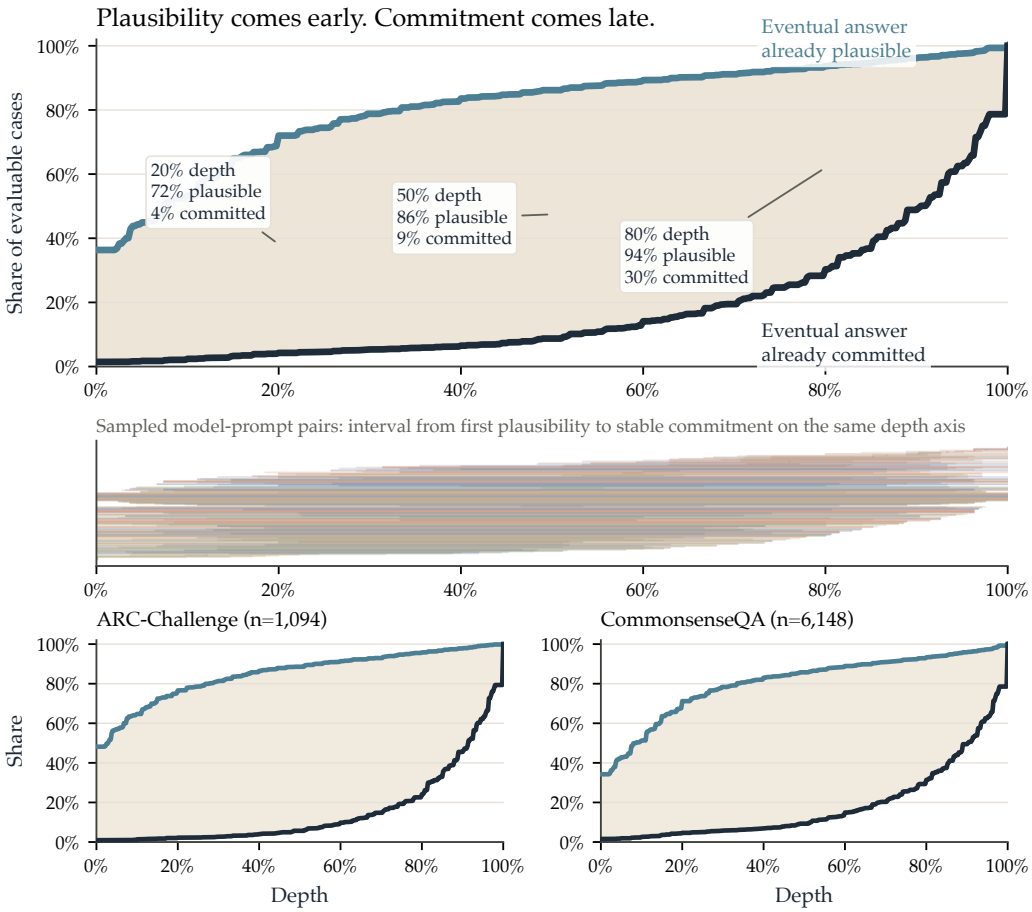


Figure 1: The eventual answer becomes plausible much earlier than it becomes committed. The top panel pools all evaluable model-prompt pairs from the primary analysis path (semantic_exact with identity ordering). The light curve asks: by this depth, in how many cases is the eventual answer already one of the plausible answers? The dark curve asks: by this depth, in how many cases has the model already opened a stable winner-runner margin that it keeps through the end of the network? The band between the curves is the main object in the paper: cases where the answer already looks plausible but the model has not fully committed yet. The interval strip shows a sample of individual cases on the same normalized-depth axis, and the two bottom panels show the same pattern separately for ARC-Challenge and CommonsenseQA.

senseQA cell. Third, it shows that the same broad lag remains visible under two control answer formats, even though those controls change coverage and can change the final winner.

2 Setup and measurement

Table 1 summarizes the experiment design. The model panel contains nine instruction-tuned checkpoints from three families: Qwen 2.5 Instruct (Qwen Team, 2024), Gemma 2 IT (Google, 2024), and Llama Instruct (AI@Meta, 2024). The reported benchmark slice contains 195 ARC-Challenge prompts (Clark et al., 2018) and 794 CommonsenseQA prompts (Talmor et al., 2019).

For every model, we render a normal multiple-choice prompt and end it with the literal suffix “Final answer:”. After that suffix, we score three possible continuations:

Aspect	Configuration
Models	Nine instruction-tuned checkpoints from Qwen 2.5 Instruct (1.5B, 3B, 7B), Gemma 2 IT (2B, 9B, 27B), and Llama Instruct (3.2 1B, 3.2 3B, 3.1 8B).
Dataset slice	ARC-Challenge (195) and CommonsenseQA (794).
Prompting	Model-specific chat or raw prompt templates. Every prompt ends with the literal suffix "Final answer:".
Readouts	Primary: the exact option text after "Final answer:", for example "photosynthesis". Controls: the literal continuation "The answer is {option_text}" and the bare answer letter, such as "A".
Permutations	Identity ordering and reverse ordering, kept paired on the same example subset.
Primary rule	For each model-dataset cell, report the median commitment lag $\Delta_{\text{margin}} = d_{\text{margin}} - d_{\text{id}}$ with a grouped-bootstrap 95% confidence interval; the cell is positive iff the lower confidence bound is greater than zero.

Table 1: Reader-facing summary of the experiment design. The main question stays fixed throughout the paper: scoring the exact option text is primary, the two alternative continuation strings are controls, and identity versus reverse answer order is kept paired on the same example subset.

- the exact option text itself, such as "photosynthesis";
- the literal template "The answer is {option_text}";
- the bare answer letter, such as "A".

The exact option-text path is the preregistered primary analysis. The other two are controls. They keep the same prompt and the same answer choices, but they change which exact continuation string we score after "Final answer:".

We score these continuations under teacher forcing, one layer at a time. In practical terms, we do not let the model free-generate a long answer. We stop at "Final answer:" and ask how much probability the network assigns to each allowed continuation at each layer. Let $p_\ell(o)$ denote the normalized probability of option o at layer ℓ. We sort options by $p_\ell(o)$ and define a plausible answer set C_ℓ as the smallest prefix whose cumulative probability reaches $\tau = 0.8$. If the cutoff lands inside a numerical tie, we mark that layer as ambiguous instead of forcing an arbitrary choice. In words, C_ℓ is the small set of answers that the model still treats as seriously possible at layer ℓ.

Our primary anchor is the eventual answer, meaning the option that wins at the final layer under the primary readout. Denote it by $o^\star$. We then track three depth events:

$$d_{\text{id}} = \min\{\ell : o^\star \in C_\ell\}, \tag{1}$$

$$d_{\text{top1}} = \min\{\ell : o^\star \text{ is the unique top-1 option at every later layer}\}, \tag{2}$$

$$d_{\text{margin}} = \min\{\ell : p_t(o^\star) - p_t(o_{(2)}) \geq \delta \text{ for all } t \geq \ell\}, \tag{3}$$

where $o_{(2)}$ is the runner-up option and $\delta = 0.15$. In words, d_{id} is when the eventual answer first enters the plausible set, d_{top1} is when it becomes the unique top prediction and stays there, and d_{margin} is when it opens a stable margin over the runner-up and never loses it.

The paper's main depth-gap statistic is

$$\Delta_{\text{margin}} = d_{\text{margin}} - d_{\text{id}}. \tag{4}$$

This quantity asks a simple question: once the eventual answer is already in the plausible set, how many layers does the model still take before it clearly commits? A case is evaluable for the primary statistic when both d_{id} and d_{margin} are defined.

The preregistered primary rule is applied separately to each model-dataset cell on the primary path (semantic_exact, identity ordering). For each cell we report the median of Δ_{margin} over valid examples and estimate a grouped-bootstrap 95% confidence interval by resampling example IDs. A cell is positive if the observed median is greater than zero and the lower confidence bound is also greater than zero. We also report two additional summaries. The first is a robustness check based on the fraction of valid examples with positive

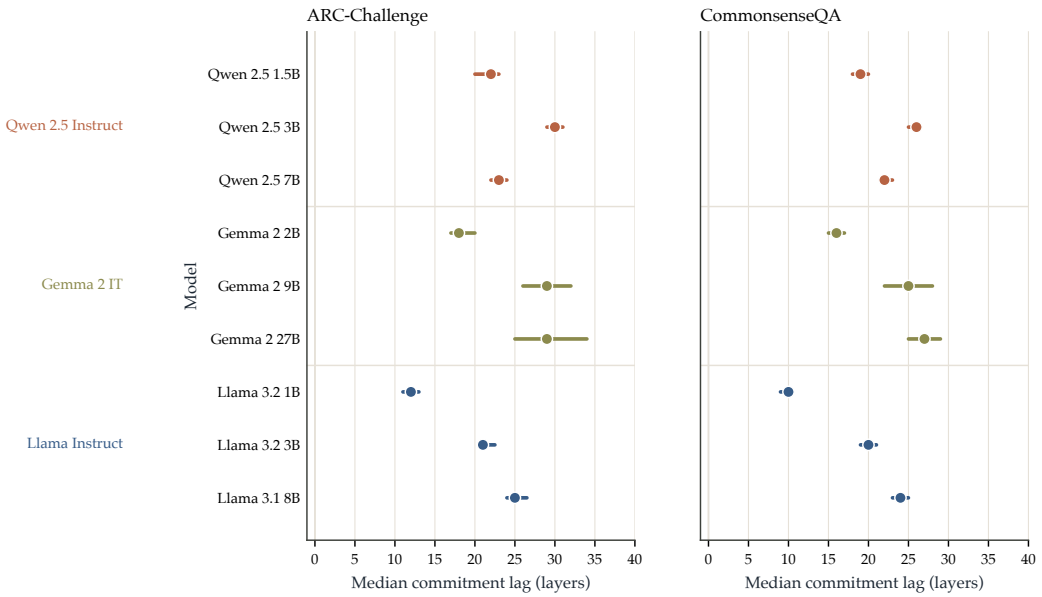


Figure 2: Every evaluable ARC-Challenge and CommonsenseQA cell is positive under the preregistered primary rule. Each point is the cell-level median commitment lag Δ_{margin} in raw layer units for the primary analysis path, and each horizontal interval is a grouped-bootstrap 95% confidence interval. Every reported model-dataset cell is positive.

lag. The second repeats the analysis while anchoring on the gold answer instead of the model's final winner. These summaries help interpret the result, but they do not determine the paper's main decision rule.

This setup intentionally differs from probe-based approaches such as the tuned lens (Belrose et al., 2023). We do not train a separate probe to guess the future answer from hidden states. Instead, we keep the task fixed and score the same answer strings the model is being asked to continue with.

3 Main result: early plausibility, late commitment

Figure 1 is the central result of the paper. The two cumulative curves are far apart. By 20% of normalized depth, the eventual answer is already plausible for 72% of evaluable cases, but it is already committed for only 4%. The gap stays large through the middle of the network. At 50% of normalized depth, 86% of evaluable cases have already admitted the eventual answer into the plausible set, while only 9% have already reached stable commitment. In raw layer units, the pooled grouped-bootstrap median commitment lag is 21.0 layers with a 95% confidence interval of [20.0, 21.0].

The dataset panels show that this is not coming from just one benchmark. ARC-Challenge and CommonsenseQA both show the same broad shape: plausibility rises early, while commitment stays much lower until later in depth. So the pooled result is not an artifact of one dataset or one model family.

Figure 2 and Table 2 translate the pooled picture into the preregistered cell-level decision rule. All 18 evaluable ARC-Challenge and CommonsenseQA cells are positive, and there are no evaluated negatives. So the main pattern is not just a pooled average. It survives at the level of individual model-dataset cells.

The family-level summaries tell the same story. Qwen contributes six positive primary cells, Gemma contributes six, and Llama contributes six. So the result is broad across the chosen

Dataset	Positive	Evaluable	Insufficient
ARC-Challenge	9	9	0
CommonsenseQA	9	9	0
All reported datasets	18	18	0

Table 2: Primary-rule summary by dataset. ARC-Challenge and CommonsenseQA are positive in every evaluable model-dataset cell.

instruct-model panel, even though the exact lag size still varies with model depth and dataset difficulty.

4 Controls and robustness

The controls change only one thing: the exact string scored after "Final answer:". Figure 3 shows why that matters. Under identity ordering, the primary exact-option readout yields a 81% valid-rate for Δ_{margin}. If we keep the prompt fixed but score the literal continuation "The answer is {option}" instead, that valid-rate falls to 58%. If we score only the answer letter, it rises to 95%.

The broad phenomenon does persist under both controls. On the fully paired identity subset, the median lag is nan layers for the exact-option readout, nan layers for the templated control, and nan layers for the letter-label control. The share of positive-lag cases on that same paired subset is 78%, 58%, and 95%, respectively. So the main effect is not unique to one answer format.

The readouts also disagree on which answer the model prefers at the end. In fully paired identity cases, the templated control agrees with the exact-option path on only 60% of cases, the letter-label control agrees on only 44%, and all three readouts agree on just 31% of cases. This is exactly why we kept the controls. They tell us that the early-candidate / late-commitment pattern is fairly robust, while also showing that different answer formats can change the evaluable set and can even change the final winner. That makes the controls scientifically useful, but it also means they are not interchangeable with the primary exact-option analysis.

The answer-order control tells a different story. In Figure 3, the identity and reverse traces are nearly on top of each other. That is reassuring because it suggests the main effect is not coming from one arbitrary option order. The report-only robustness summaries also line up with the primary rule. The secondary criterion yields the same positive pattern as the primary rule (18 positive, 0 insufficient), and the gold-answer sensitivity analysis is positive on all 18 evaluable ARC-Challenge and CommonsenseQA cells.

Finally, the QC path is designed to surface problems, not hide them. The released audit bundle logs 1224 tokenization collisions and 17 duplicate-option examples instead of silently dropping them, while the merged run itself still has zero failure rows and zero unpaired rows. The point is not to make the result look tidy. The point is to show that the result still holds when we keep the pipeline honest about edge cases.

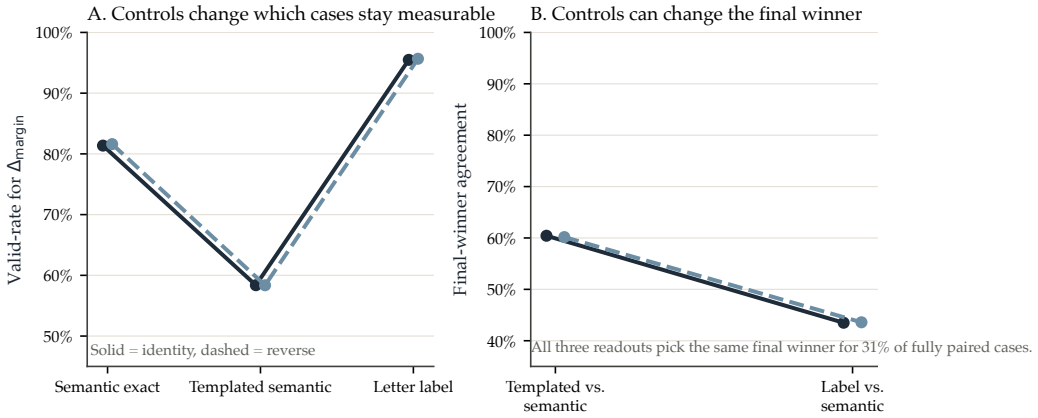


Figure 3: The broad lag survives the controls, but the controls still change the measurement. Panel A shows how often Δ_{margin} is defined for each readout. Panel B shows how often the two control readouts pick the same final winner as the semantic-exact primary readout on fully paired cases. Identity and reverse order behave similarly. The answer-format controls do not, which is why they are useful robustness checks but not replacements for the main semantic path.

5 Interpretation and limits

Figure 4 shows what the aggregate result looks like in individual cases. Each panel plots the winner-minus-runner contrast for one semantic-exact identity example. The details differ by family and prompt, but the same broad pattern appears again and again: the eventual answer becomes plausible early, while stable commitment arrives much later and can be preceded by hesitation or reversals.

These measurements still need to be interpreted carefully. We are not claiming that the model literally stores a human-like list of hypotheses, and we are not claiming that these curves identify a full causal circuit. Our claim is narrower. Under this prompting and scoring setup, the eventual answer often enters the plausible-answer set before it becomes stably isolated from the runner-up.

The empirical scope is also narrow in two practical ways. First, the paper uses a 989-question slice rather than the full benchmark splits. Second, the panel contains only instruction-tuned open-weight models, so the claim should not be generalized to base models or closed systems.

Those limits matter, but they do not erase the signal. The early-plausibility/late-commitment pattern appears in the pooled depth curves, in every evaluable ARC-Challenge and CommonsenseQA cell, and in the prompt-level trajectories. The controls matter here too. They show that small changes in answer format can change coverage and final winners, which is exactly why the paper keeps exact option-text scoring as the main path.

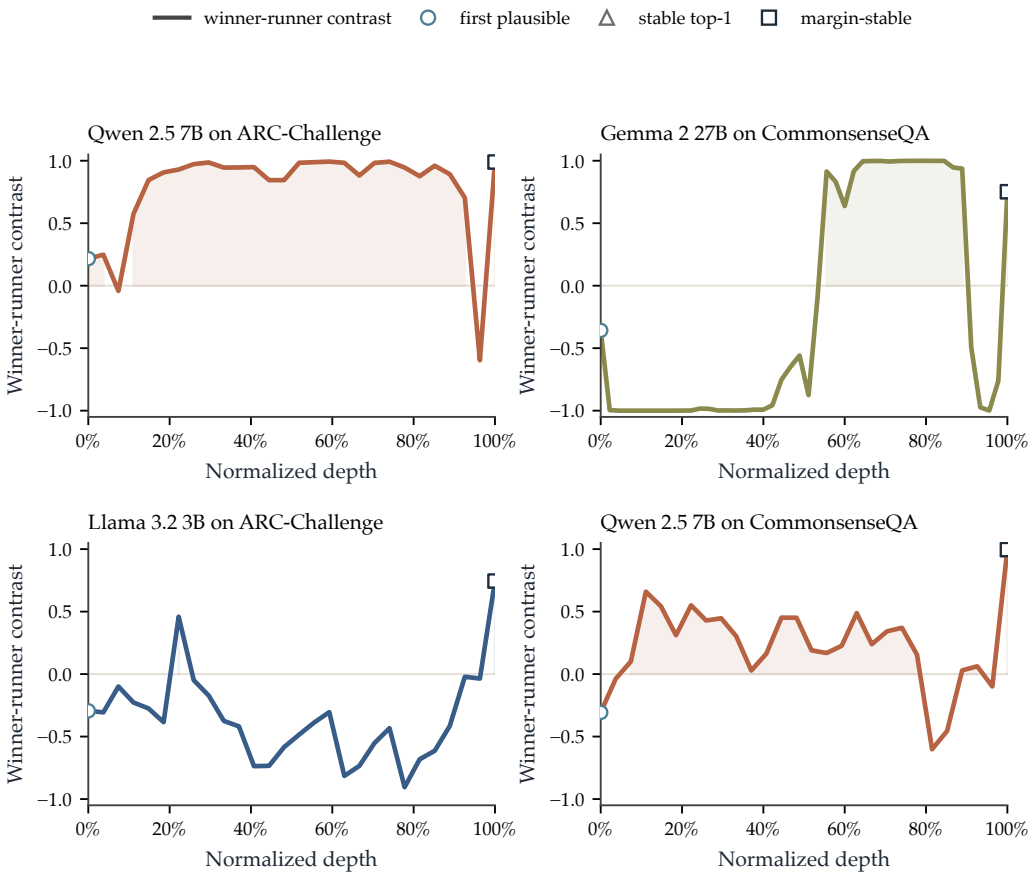


Figure 4: Prompt-level trajectories show the same pattern as the pooled figures. Each panel plots the winner-minus-runner contrast across normalized depth for one semantic-exact identity example. The circular marker shows when the eventual answer first enters the plausible set, the triangular marker shows when it becomes stable top-1, and the square marker shows when it becomes margin-stably committed. The exact shapes vary, but the same left-to-right separation between plausibility and commitment appears at the single-case level.

6 Conclusion

The paper's claim is simple and deliberately scoped. In the setting we study, instruction-tuned models often know which answers are still possible before they have fully settled on the one they will finally say. We show that pattern by ending each prompt with "Final answer:" and tracking how the scores of exact answer continuations change across model depth.

The evidence is consistent across levels of analysis. The flagship pooled curves show a large gap between plausibility and commitment. Every evaluable ARC-Challenge and CommonsenseQA model-dataset cell is positive under the preregistered primary rule. The prompt-level examples show the same separation in concrete trajectories. The answer-format controls matter because they show that this result is not automatic: changing the continuation string can change both coverage and winners.

The result does not yet explain why the gap appears, and it should not be overgeneralized beyond this panel and benchmark slice. But it does support a useful empirical distinction for future work. The moment when a model finally says an answer can come much later

than the moment when that answer first becomes one of the answers the model already treats as plausible.

References

AI@Meta. The Llama 3 herd of models. arXiv preprint arXiv:2407.21783, 2024.

Nora Belrose, Zach Furman, Logan Smith, Danny Halawi, Igor Ostrovsky, Lev McKinney, Stella Biderman, and Jacob Steinhardt. Eliciting latent predictions from transformers with the tuned lens. arXiv preprint arXiv:2303.08112, 2023.

Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? try ARC, the AI2 reasoning challenge. arXiv preprint arXiv:1803.05457, 2018.

Google. Gemma 2 instruction-tuned model family. https://huggingface.co/google/gemma-2-9b-it, 2024. Accessed March 30, 2026.

Qwen Team. Qwen2.5 Instruct model family. https://huggingface.co/Qwen/Qwen2.5-7B-Instruct, 2024. Accessed March 30, 2026.

Alon Talmor, Jonathan Herzig, Nicholas Lourie, and Jonathan Berant. CommonsenseQA: A question answering challenge targeting commonsense knowledge. In Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics, pp. 4149–4158, 2019.

A Supplementary material

A.1 Reproducibility statement

This manuscript is generated from a single merged run, paper_merged_9x1000_20260330T122258Z, with run-scoped manifests, QC summaries, and figure/table provenance checked into the paper workspace. The reported analysis uses the ARC-Challenge and CommonsenseQA portion of that run. All numeric claims in the paper are generated from those artifacts rather than entered by hand.

Ethics statement

This study analyzes public benchmark questions using open-weight language models and does not involve human subjects, personal data, or deployment to end users. The main risk is analytical overclaiming: layerwise statistics can be mistaken for a complete mechanistic explanation when they are only an operational measurement. We mitigate that risk by keeping the claim narrow, preregistering the primary rule, preserving anomaly logs, and distinguishing the primary exact-option analysis from the control readouts.

AI-use disclosure

AI-use disclosure. The authors used AI-assisted tools during the project lifecycle: ChatGPT for brainstorming and experimental design iteration, Codex for implementation assistance, and Claude Opus for drafting and formatting portions of the manuscript and LaTeX source. All scientific decisions, experiments, analyses, code review, result verification, and final manuscript editing were performed and approved by the human authors, who take full responsibility for the paper's contents.

A.2 Model registry

Family	Model	Size	Repository	Revision prefix
Qwen 2.5 Instruct	Qwen 2.5 1.5B	1.5b	Qwen2.5-1.5B-Instruct	989aa7980e4c
Qwen 2.5 Instruct	Qwen 2.5 3B	3b	Qwen2.5-3B-Instruct	aa8e72537993
Qwen 2.5 Instruct	Qwen 2.5 7B	7b	Qwen2.5-7B-Instruct	a09a35458c70
Gemma 2 IT	Gemma 2 2B	2b	gemma-2-2b-it	299a8560bedf
Gemma 2 IT	Gemma 2 9B	9b	gemma-2-9b-it	11c9b309abf7
Gemma 2 IT	Gemma 2 27B	27b	gemma-2-27b-it	aaf20e6b9f4c
Llama Instruct	Llama 3.2 1B	1b	Llama-3.2-1B-Instruct	9213176726f5
Llama Instruct	Llama 3.2 3B	3b	Llama-3.2-3B-Instruct	0cb88a4f764b
Llama Instruct	Llama 3.1 8B	8b	Meta-Llama-3.1-8B-Instruct	0e9e39f249a1

Table 3: Model registry used in the paper. Short repository names and 12-character revision prefixes are shown for readability; the full pinned revision hashes are preserved in the released provenance manifest.

A.3 Full primary positivity table

Dataset	Model	Valid / total	Median lag	95% CI	Primary decision
ARC-Challenge	Qwen 2.5 1.5B	109/195	22	[20, 23]	positive
ARC-Challenge	Qwen 2.5 3B	143/195	30	[29, 31]	positive
ARC-Challenge	Qwen 2.5 7B	163/195	23	[22, 24]	positive
ARC-Challenge	Gemma 2 2B	117/195	18	[17, 20]	positive
ARC-Challenge	Gemma 2 9B	132/195	29	[26, 32]	positive
ARC-Challenge	Gemma 2 27B	135/195	29	[25, 34]	positive
ARC-Challenge	Llama 3.2 1B	79/195	12	[11, 13]	positive
ARC-Challenge	Llama 3.2 3B	104/195	21	[20, 22]	positive
ARC-Challenge	Llama 3.1 8B	112/195	25	[24, 26]	positive
CommonsenseQA	Qwen 2.5 1.5B	677/794	19	[18, 20]	positive
CommonsenseQA	Qwen 2.5 3B	706/794	26	[25, 26]	positive
CommonsenseQA	Qwen 2.5 7B	718/794	22	[22, 23]	positive
CommonsenseQA	Gemma 2 2B	693/794	16	[15, 17]	positive
CommonsenseQA	Gemma 2 9B	702/794	25	[22, 28]	positive
CommonsenseQA	Gemma 2 27B	706/794	27	[25, 29]	positive
CommonsenseQA	Llama 3.2 1B	625/794	10	[9, 10]	positive
CommonsenseQA	Llama 3.2 3B	672/794	20	[19, 21]	positive
CommonsenseQA	Llama 3.1 8B	649/794	24	[23, 25]	positive

Table 4: Full cell-level primary-rule results. The primary decision is evaluated separately for each model-dataset cell on the semantic-exact identity path.

A.4 Selection and QC details

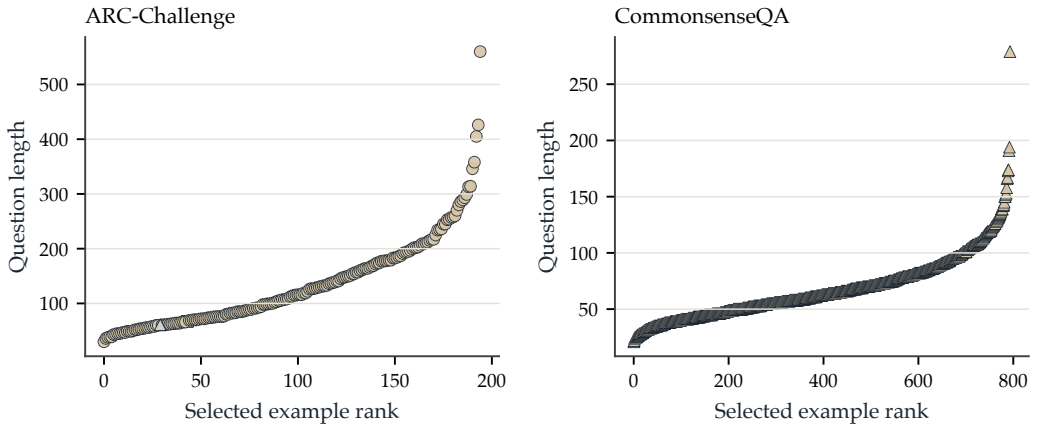


Figure 5: The paper slice spans a wide range of selected prompt lengths. The figure shows the selected-example question-length profile for each reported dataset together with the chosen option-count markers.

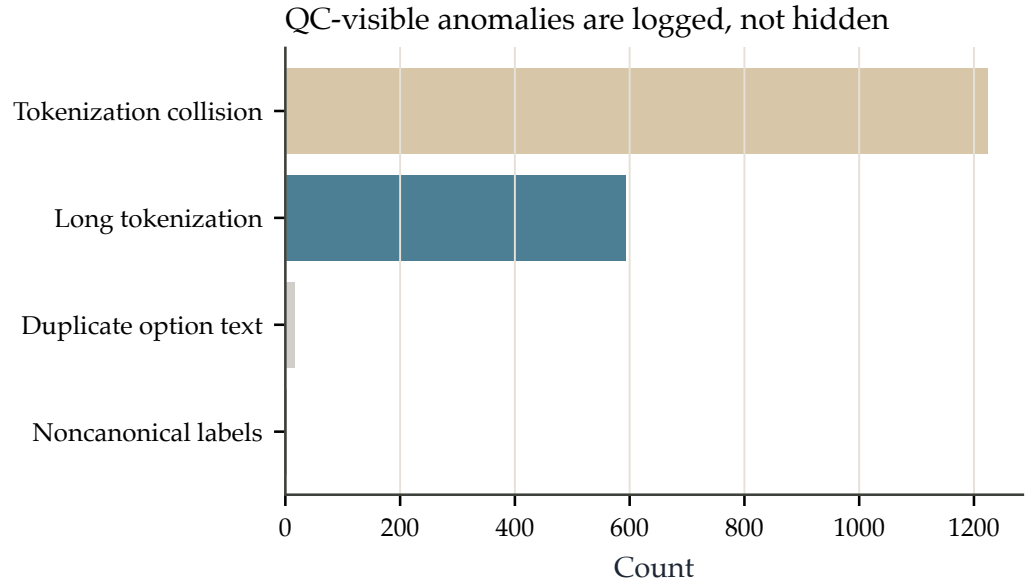


Figure 6: QC-visible anomalies are logged rather than hidden. These counts come directly from the merged run's QC summary. They motivate caution, but they do not arise from silent exclusion or failed pairing.

QC item	Count
Prepared examples	1000
Token audits	258876
Tokenization collisions	1224
Suspiciously long option tokenizations	594
Duplicate option text	17
Noncanonical option labels	2
Unpaired rows	0
Failure rows	0

Table 5: QC summary for the merged run. The run completes with zero failure rows and zero unpaired rows while still surfacing tokenization and option-surface anomalies.

Readout	Continuation surface	Normalization
semantic_exact	Exact option text after "Final answer:", e.g. "photosynthesis"	Mean log-probability
templated_semantic	Literal continuation "The answer is {option_text}"	Mean log-probability
letter_label	Bare answer letter after "Final answer:", e.g. "A"	Mean log-probability

Table 6: Readout definitions used in the paper. The controls intentionally differ in continuation surface while keeping the rest of the setup fixed.